

Exercises and Problems for Section 13.4

Exercises

In Exercises 1–7, use the algebraic definition to find $\vec{v} \times \vec{w}$.

1. $\vec{v} = \vec{k}, \vec{w} = \vec{j}$
2. $\vec{v} = -\vec{i}, \vec{w} = \vec{j} + \vec{k}$
3. $\vec{v} = \vec{i} + \vec{k}, \vec{w} = \vec{i} + \vec{j}$
4. $\vec{v} = \vec{i} + \vec{j} + \vec{k}, \vec{w} = \vec{i} + \vec{j} - \vec{k}$
5. $\vec{v} = 2\vec{i} - 3\vec{j} + \vec{k}, \vec{w} = \vec{i} + 2\vec{j} - \vec{k}$
6. $\vec{v} = 2\vec{i} - \vec{j} - \vec{k}, \vec{w} = -6\vec{i} + 3\vec{j} + 3\vec{k}$
7. $\vec{v} = -3\vec{i} + 5\vec{j} + 4\vec{k}, \vec{w} = \vec{i} - 3\vec{j} - \vec{k}$

Use the geometric definition in Exercises 8–9 to find:

8. $2\vec{i} \times (\vec{i} + \vec{j})$
9. $(\vec{i} + \vec{j}) \times (\vec{i} - \vec{j})$

In Exercises 10–11, use the properties on page 746 to find:

10. $((\vec{i} + \vec{j}) \times \vec{i}) \times \vec{j}$
11. $(\vec{i} + \vec{j}) \times (\vec{i} \times \vec{j})$

12. For $\vec{a} = 3\vec{i} + \vec{j} - \vec{k}$ and $\vec{b} = \vec{i} - 4\vec{j} + 2\vec{k}$, find $\vec{a} \times \vec{b}$ and check that it is perpendicular to both $\vec{a}$ and $\vec{b}$.
13. If $\vec{v} = 3\vec{i} - 2\vec{j} + 4\vec{k}$ and $\vec{w} = \vec{i} + 2\vec{j} - \vec{k}$, find $\vec{v} \times \vec{w}$ and $\vec{w} \times \vec{v}$. What is the relation between the two answers?

In Exercises 14–15, find an equation for the plane through the points.

14. $(1, 0, 0), (0, 1, 0), (0, 0, 1)$.
15. $(3, 4, 2), (-2, 1, 0), (0, 2, 1)$.

In Exercises 16–19, find the volume of the parallelepiped with edges $\vec{a}, \vec{b}, \vec{c}$.

16. $\vec{a} = 3\vec{i} + 4\vec{j} + 5\vec{k}, \vec{b} = 5\vec{i} + 4\vec{j} + 3\vec{k}, \vec{c} = \vec{i} + \vec{j} + \vec{k}$.
17. $\vec{a} = -\vec{i} + \vec{j} + \vec{k}, \vec{b} = \vec{i} - \vec{j} + \vec{k}, \vec{c} = \vec{i} + \vec{j} - \vec{k}$.
18. $\vec{a} = -\vec{i} + 8\vec{j} + 7\vec{k}, \vec{b} = 2\vec{j} + 9\vec{k}, \vec{c} = 3\vec{k}$.
19. $\vec{a} = \vec{i} + \vec{j} + 2\vec{k}, \vec{b} = \vec{i} + \vec{k}, \vec{c} = \vec{j} + \vec{k}$.

Problems

20. Find a vector parallel to the line of intersection of the planes given by the equations $2x - 3y + 5z = 2$ and $4x + y - 3z = 7$.
21. Find the equation of the plane through the origin that is perpendicular to the line of intersection of the planes in Problem 20.
22. Find the equation of the plane through the point $(4, 5, 6)$ and perpendicular to the line of intersection of the planes in Problem 20.
23. Find an equation for the plane through the origin containing the points $(1, 3, 0)$ and $(2, 4, 1)$.
24. Find a vector parallel to the line of intersection of the two planes $4x - 3y + 2z = 12$ and $x + 5y - z = 25$.
25. Find a vector parallel to the intersection of the planes $2x - 3y + 5z = 2$ and $4x + y - 3z = 7$.
26. Find the equation of the plane through the origin that is perpendicular to the line of intersection of the planes in Problem 25.
27. Find the equation of the plane through the point $(4, 5, 6)$ that is perpendicular to the line of intersection of the planes in Problem 25.
28. Find the equation of a plane through the origin and perpendicular to $x - y + z = 5$ and $2x + y - 2z = 7$.
29. Given the points $P = (1, 2, 3)$, $Q = (3, 5, 7)$, and $R = (2, 5, 3)$, find:
 - (a) A unit vector perpendicular to a plane containing P, Q, R .
 - (b) The angle between PQ and PR .
 - (c) The area of the triangle PQR .
 - (d) The distance from R to the line through P and Q .
30. Let $A = (-1, 3, 0)$, $B = (3, 2, 4)$, and $C = (1, -1, 5)$.
 - (a) Find an equation for the plane that passes through these three points.
 - (b) Find the area of the triangle determined by these three points.
31. If $\vec{v}$ and $\vec{w}$ are both parallel to the xy -plane, what can you conclude about $\vec{v} \times \vec{w}$? Explain.
32. Suppose $\vec{v} \cdot \vec{w} = 5$ and $\|\vec{v} \times \vec{w}\| = 3$, and the angle between $\vec{v}$ and $\vec{w}$ is θ . Find
 - (a) $\tan \theta$
 - (b) θ .
33. If $\vec{v} \times \vec{w} = 2\vec{i} - 3\vec{j} + 5\vec{k}$, and $\vec{v} \cdot \vec{w} = 3$, find $\tan \theta$ where θ is the angle between $\vec{v}$ and $\vec{w}$.
34. Suppose $\vec{v} \cdot \vec{w} = 8$ and $\vec{v} \times \vec{w} = 12\vec{i} - 3\vec{j} + 4\vec{k}$ and that the angle between $\vec{v}$ and $\vec{w}$ is θ . Find
 - (a) $\tan \theta$
 - (b) θ
35. Why does a baseball curve? The baseball in Figure 13.42 has velocity $\vec{v}$ meters/sec and is spinning at ω radians per second about an axis in the direction of the unit vector $\vec{n}$. The ball experiences a force, called the Magnus force,² $\vec{F}_M$, that is proportional to $\omega \vec{n} \times \vec{v}$.

²Named after German physicist Heinrich Magnus, who first described it in 1853.

- (a) What is the effect on $\vec{F}_M$ of increasing ω ?
 (b) The ball in Figure 13.42 is moving away from you. What is the direction of the Magnus force?



Figure 13.42: Spinning baseball

36. The point P in Figure 13.43 has position vector $\vec{v}$ obtained by rotating the position vector $\vec{r}$ of the point (x, y) by 90° counterclockwise about the origin.

- (a) Use the geometric definition of the cross product to explain why $\vec{v} = \vec{k} \times \vec{r}$.
 (b) Find the coordinates of P .

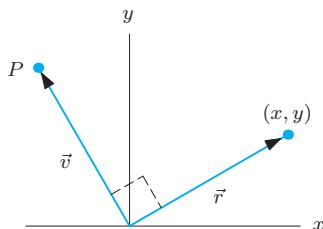


Figure 13.43

37. The points $P_1 = (0, 0, 0)$, $P_2 = (2, 4, 2)$, $P_3 = (3, 0, 0)$, and $P_4 = (5, 4, 2)$ are vertices of a parallelogram.

- (a) Find the displacement vectors along each of the four sides. Check that these are equal in pairs.
 (b) Find the area of the parallelogram.

38. Using the parallelogram in Problem 37 as a base, create a parallelepiped with side $\overrightarrow{P_1P_5}$ where $P_5 = (1, 0, 4)$. Find the volume of this parallelepiped.

39. Use the algebraic definition to check that

$$\vec{a} \times (\vec{b} + \vec{c}) = (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c}).$$

40. If $\vec{v}$ and $\vec{w}$ are non-zero vectors, use the geometric definition of the cross product to explain why

$$(\lambda \vec{v}) \times \vec{w} = \lambda(\vec{v} \times \vec{w}) = \vec{v} \times (\lambda \vec{w}).$$

Consider the cases $\lambda > 0$, and $\lambda = 0$, and $\lambda < 0$ separately.

41. Use a parallelepiped to show that $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$ for any vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$.

42. Show that $\|\vec{a} \times \vec{b}\|^2 = \|\vec{a}\|^2 \|\vec{b}\|^2 - (\vec{a} \cdot \vec{b})^2$.

43. If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, show that

$$\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}.$$

Geometrically, what does this imply about $\vec{a}$, $\vec{b}$, and $\vec{c}$?

44. If $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ and $\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$ are any three vectors in space, show that

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

45. Use the fact that $\vec{i} \times \vec{i} = \vec{0}$, $\vec{i} \times \vec{j} = \vec{k}$, $\vec{i} \times \vec{k} = -\vec{j}$, and so on, together with the properties on page 746 to derive the algebraic definition for the cross product.

46. In this problem, we arrive at the algebraic definition for the cross product by a different route. Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$. We seek a vector $\vec{v} = x\vec{i} + y\vec{j} + z\vec{k}$ that is perpendicular to both $\vec{a}$ and $\vec{b}$. Use this requirement to construct two equations for x , y , and z . Eliminate x and solve for y in terms of z . Then eliminate y and solve for x in terms of z . Since z can be any value whatsoever (the direction of $\vec{v}$ is unaffected), select the value for z which eliminates the denominator in the equation you obtained. How does the resulting expression for $\vec{v}$ compare to the formula we derived on page 745?

47. For vectors $\vec{a}$ and $\vec{b}$, let $\vec{c} = \vec{a} \times (\vec{b} \times \vec{a})$.

- (a) Show that $\vec{c}$ lies in the plane containing $\vec{a}$ and $\vec{b}$.
 (b) Use Problems 41 and 42 to show that $\vec{a} \cdot \vec{c} = 0$ and $\vec{b} \cdot \vec{c} = \|\vec{a}\|^2 \|\vec{b}\|^2 - (\vec{a} \cdot \vec{b})^2$.
 (c) Show that

$$\vec{a} \times (\vec{b} \times \vec{a}) = \|\vec{a}\|^2 \vec{b} - (\vec{a} \cdot \vec{b}) \vec{a}.$$

48. Use the result of Problem 41 to show that the cross product distributes over addition. First, use distributivity for the dot product to show that for any vector $\vec{d}$,

$$[(\vec{a} + \vec{b}) \times \vec{c}] \cdot \vec{d} = [(\vec{a} \times \vec{c}) + (\vec{b} \times \vec{c})] \cdot \vec{d}.$$

Next, show that for any vector $\vec{d}$,

$$[((\vec{a} + \vec{b}) \times \vec{c}) - (\vec{a} \times \vec{c}) - (\vec{b} \times \vec{c})] \cdot \vec{d} = 0.$$

Finally, explain why you can conclude that

$$(\vec{a} + \vec{b}) \times \vec{c} = (\vec{a} \times \vec{c}) + (\vec{b} \times \vec{c}).$$

49. Figure 13.44 shows the tetrahedron determined by three vectors $\vec{a}$, $\vec{b}$, $\vec{c}$. The *area vector* of a face is a vector perpendicular to the face, pointing outward, whose magnitude is the area of the face. Show that the sum of the four outward-pointing area vectors of the faces equals the zero vector.

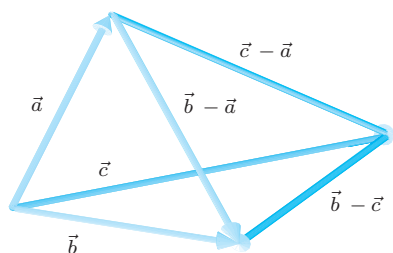


Figure 13.44

In Problems 50–52, find the vector representing the area of a surface. The magnitude of the vector equals the magnitude of the area; the direction is perpendicular to the surface. Since there are two perpendicular directions, we pick one by giving an orientation for the surface.

50. The rectangle with vertices $(0, 0, 0)$, $(0, 1, 0)$, $(2, 1, 0)$, and $(2, 0, 0)$, oriented so that it faces downward.
51. The circle of radius 2 in the yz -plane, facing in the direction of the positive x -axis.
52. The triangle ABC , oriented upward, where $A = (1, 2, 3)$, $B = (3, 1, 2)$, and $C = (2, 1, 3)$.
53. This problem relates the area of a parallelogram S lying in the plane $z = mx + ny + c$ to the area of its projection

R in the xy -plane. Let S be determined by the vectors $\vec{u} = u_1\vec{i} + u_2\vec{j} + u_3\vec{k}$ and $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$. See Figure 13.45.

- (a) Find the area of S .
- (b) Find the area of R .
- (c) Find m and n in terms of the components of $\vec{u}$ and $\vec{v}$.
- (d) Show that

$$\text{Area of } S = \sqrt{1 + m^2 + n^2} \cdot \text{Area of } R.$$

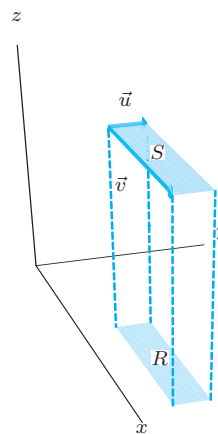


Figure 13.45

Strengthen Your Understanding

In Problems 54–55, explain what is wrong with the statement.

54. There is only one unit vector perpendicular to two non-parallel vectors in 3-space.
55. $\vec{u} \times \vec{v} = \vec{0}$ when $\vec{u}$ and $\vec{v}$ are perpendicular.

In Problems 56–57, give an example of:

56. A vector $\vec{u}$ whose cross product with $\vec{v} = \vec{i} + \vec{j}$ is parallel to $\vec{k}$.
57. A vector $\vec{v}$ such that $\|\vec{u} \times \vec{v}\| = 10$, where $\vec{u} = 3\vec{i} + 4\vec{j}$.

Are the statements in Problems 58–67 true or false? Give reasons for your answer.

58. $\vec{u} \times \vec{v}$ is a vector.

59. $\vec{u} \times \vec{v}$ has direction parallel to both $\vec{u}$ and $\vec{v}$.

60. $\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \|\vec{v}\|$.

61. $(\vec{i} \times \vec{j}) \cdot \vec{k} = \vec{i} \cdot (\vec{j} \times \vec{k})$.

62. If $\vec{v}$ is a non-zero vector and $\vec{v} \times \vec{u} = \vec{v} \times \vec{w}$, then $\vec{u} = \vec{w}$.

63. The value of $\vec{v} \cdot (\vec{v} \times \vec{w})$ is always 0.

64. The value of $\vec{v} \times \vec{w}$ is never the same as $\vec{v} \cdot \vec{w}$.

65. The area of the triangle with two sides given by $\vec{i} + \vec{j}$ and $\vec{j} + 2\vec{k}$ is $3/2$.

66. Given a non-zero vector $\vec{v}$ in 3-space, there is a non-zero vector $\vec{w}$ such that $\vec{v} \times \vec{w} = \vec{0}$.

67. It is never true that $\vec{v} \times \vec{w} = \vec{w} \times \vec{v}$.